

Dexter Fernandes

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Computer Vision Engineer

Computer Vision Engineer with 3.5 years' experience designing and deploying real-time edge-AI systems in high-stakes environments (oil rigs, industrial sites). Proven track record reducing inference latency to sub-50 ms and boosting detection accuracy by up to 33%. Seeking to lead perception algorithm development for space-based autonomy.

WORK EXPERIENCE

Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Developed time-series machine learning models for the classification of on-shore oil rig operations to increase worker safety and reduce supervision required during 24/7 operation, achieving a 23% higher F1 score than human experts.
- Spearheaded the end-to-end development and production rollout of YOLOv5 models, to elevate rig worker and equipment detection precision by 15% in 3 months.
- Developed a tool for collecting raw sensor data and applying automated augmentation and labeling workflows to boost dataset diversity by 40% and cut manual annotation effort by 60%.
- Led strategic MLOps initiatives to continuously enhance experimentation and deployment workflows through bayesian hyperparameter tuning of time-series and vision models.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Refactored a legacy monolithic system into a Kafka-driven microservice architecture, containerising each service with Docker to enhance failure isolation and improve MTBF by 35% within four months
- Designed a dynamic memory management solution to address recurring crashes in a live video streaming system, reducing crash frequency by 67%.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-engineer team on ViCAS to design, train, and deploy deep learning models for real-time object detection, pose estimation, and tracking on edge hardware, enhancing model accuracy and reducing false positives by 33% while achieving sub-50 ms inference latency within 6 months.
- Developed and deployed core perception algorithms on resource-constrained Intel edge devices, leveraging mixed-precision quantization and structured pruning to enable real-time decision-making in harsh industrial environments.
- Implemented real-time object tracking with YOLO-DeepSORT, maintaining consistent identities across frames and enabling reliable downstream analytics.
- Optimized neural network architectures using quantization-aware training, structured pruning, and layer fusion thus reducing model size by 50%.
- Implemented rigorous testing and validation frameworks based on simulated adverse conditions, to ensure model accuracy and stability, reducing performance variance by 25% under challenging lighting, occlusion, and noise scenarios.
- Engineered synthetic training data pipelines using OpenCV and NumPy, reducing edge-case model development time from 6 days to 2 days for high-risk production environments.

EDUCATION

Master of Science in Robotics - 2:1

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

- Engineered a UGV system (Clearpath Husky) with multi-sensor fusion (Velodyne LiDAR, Intel RealSense RGBD camera, Symetrica Verifinder).
- Deployed real-time LiDAR SLAM (ROS2 Nav2 + Velodyne) on Jetson AGX Xavier, containerised with Docker for field-deployable, point-cloud mapping.
- Achieved precise radionuclide localisation using advanced odometry techniques like Scan-to-Map ICP, by significantly reducing drift in real-time LiDAR SLAM operations.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

SKILLS

- **Perception (Core):** LiDAR SLAM, Nav2, OpenCV, OpenVINO, ROS2, Vision Transformer (ViT), YOLOv5/v8
- **Perception (Supporting):** Computer Vision Annotation Tool (CVAT), Neural Network Compression Framework (NNCF)
- **Backend & Cloud / Edge:** Apache Kafka, AWS, FastAPI, Intel VPU, Microsoft Azure, NVIDIA Jetson AGX, PostgreSQL
- **DevOps & MLOps:** Docker, DVC, Git, Linux, MLFlow, Optuna